

A Minor Project Report

On

CYBERSECURITY AWARENESS TRAINING WEBSITE

(Interactive Web-Based Cybersecurity Education Platform)

Developed By

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as

**Partial Fulfillment of Semester V of
Bachelor of Science (Information Technology) Honours
Division D**

For A.Y. 2026 – 2027

Under the Guidance of

Prof. Sohil Parmar

Submitted To

Department of B.Sc. IT
Faculty of IT & Computer Science

Parul University

Certificate

This is to certify that **Dhairya Kahar** (2405101020009), student of B.Sc. (Information Technology) Honours, Division D, has satisfactorily completed the Minor Project titled "**Cybersecurity Awareness Training Website**" as partial fulfillment of Semester V, Faculty of IT & Computer Science, Parul University.

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Acknowledgement

We express our sincere gratitude to the Faculty of IT & Computer Science, Parul University, for the opportunity to undertake this Minor Project as part of our B.Sc. (IT) Honours Semester V curriculum.

We are especially thankful to our project guide, **Prof. Sohil Parmar**, for continuous guidance, valuable feedback, and encouragement throughout the design and development of this project. His insightful technical critique and direction were invaluable in shaping the architecture and implementation of the Cybersecurity Awareness Training Website.

We also thank the faculty members of the B.Sc. IT department and the laboratory staff for their support, and our families and peers for their constant motivation and understanding during the course of this work.

Dhairya Kahar

Contents

List of Figures	v
List of Tables	vi
I Introduction & Project Profile	1
1 About the Institution	2
1.1 Parul University	2
1.2 Faculty of IT & Computer Science	2
1.3 Department of B.Sc. Information Technology	2
2 Project Profile	3
2.1 Project Definition	3
2.2 Project Description	3
2.3 Existing System	3
2.4 Problem Statement	3
2.5 Need for the New System	4
2.6 Proposed System & Key Features	4
2.7 Scope	4
2.8 Expected Outcomes	4
2.9 Tools & Technology Used	4
II Analysis & Design	5
3 Requirement Analysis	6
3.1 Feasibility Study	6
3.2 Users of the System	6
3.3 Functional Requirements	6
3.4 Non-Functional Requirements	7
3.5 Hardware & Software Requirements	7
4 System Design	8
4.1 Use Case Diagram	8

Part I

Introduction & Project Profile

1 About the Institution

1.1 Parul University

Parul University is established under the Gujarat Private University Act, 2009, following legislation passed by the Government of Gujarat on 26th March 2015, granting university status to the Parul Group of Institutes functioning under Parul Arogya Seva Mandal Trust. The University has emerged as a premier multidisciplinary institution in western India, imparting world-class technical education, fostering entrepreneurial ecosystems, and driving impactful technological innovations across engineering, applied sciences, and computer applications.

1.2 Faculty of IT & Computer Science

The Faculty of IT & Computer Science (FITCS) is one of the leading IT education providers in the region, preparing industry-ready professionals through a strong blend of technical and practical learning across its B.Sc. IT, B.Sc. IT Honours, BCA, IMCA and MCA programs. The faculty emphasizes modern curricula aligned with contemporary software engineering paradigms, full-stack application development, distributed architectures, and agile engineering practices.

1.3 Department of B.Sc. Information Technology

The B.Sc. IT department focuses on building strong programming, systems, and database fundamentals, encouraging students to apply classroom learning to real, deployable software through semester-wise minor and major projects. Through project-centric pedagogy, students engineer robust digital solutions addressing authentic societal and industrial requirements, embodying modern frameworks, robust security postures, and scalable database architectures.

2 Project Profile

2.1 Project Definition

The Cybersecurity Awareness Training Website is an interactive, full-stack web application designed to educate users on critical cybersecurity threats. It provides a structured learning environment with comprehensive topics, progress tracking, knowledge assessments via quizzes, and a practical phishing simulator to ensure users can identify and mitigate real-world cyber risks.

2.2 Project Description

The system is built on a robust Python Flask backend using a modular Blueprint architecture, integrated with a relational database (PostgreSQL/SQLite) via SQLAlchemy ORM. The frontend utilizes HTML5,

2.4 Problem Statement

With the rapid increase in social engineering, phishing, and data breaches, average internet users and employees lack a unified, engaging platform to learn and test their cybersecurity hygiene. There is a critical need for an accessible training ecosystem that bridges the gap between theoretical security concepts and practical threat identification.

2.5 Need for the New System

- **Unified Educational Platform:** Replaces scattered resources with a structured, step-by-step curriculum.
- **Measurable Knowledge:** Enforces learning through dynamic quizzes and retains score histories.
- **Practical Phishing Simulation:** Trains users to visually identify malicious links and emails.
- **Admin Governance:** Allows instructors/admins to continually update question banks and monitor user performance.

2.6 Proposed System & Key Features

- **Multi-Role Portals:** Tailored dashboards for standard Users (Learners) and Administrators.
- **Curriculum & Progress Engine:** 8 structured cybersecurity topics with completion tracking.
- **Dynamic Quiz System:** Automated scoring, answer explanations, and percentage calculations across 40 questions.
- **Phishing Awareness Simulator:** Interactive module to test users against common phishing tactics.
- **Password Strength Analyzer:** Client-side entropy calculator and crack time estimator.
- **Admin Control Panel:** Interface to add/edit quiz questions and monitor platform metrics.
- **Secure Authentication:** Password hashing (Werkzeug) and session management (Flask-Login).

2.7 Scope

The project provides an end-to-end full-stack web application with responsive layouts, supporting user registration, learning modules, assessments, and an administrative backend, engineered for both local SQLite execution and cloud-native PostgreSQL deployment.

2.8 Expected Outcomes

A highly accessible training platform that curtails user vulnerability to cyber threats, provides measurable metrics on security awareness, and offers a scalable foundation for organizational security training.

2.9 Tools & Technology Used

- **Frontend:** HTML5, CSS3, Bootstrap 5, Jinja2 Templates, JavaScript (ES6+)
- **Backend:** Python 3.11/3.12, Flask, Flask-Login, Werkzeug

Part II

Analysis & Design

3 Requirement Analysis

3.1 Feasibility Study

Technical Feasibility: The architecture utilizes proven, lightweight frameworks. Flask provides high flexibility for web routing, while SQLAlchemy ensures database abstraction (easily switching between SQLite for dev and PostgreSQL for prod). Bootstrap guarantees UI responsiveness without heavy frontend frameworks.

Operational Feasibility: The platform addresses the clear need for security training. The intuitive dashboard ensures that non-technical users can navigate topics and quizzes with zero friction.

Economic Feasibility: The system is built entirely on open-source technologies (Python, Flask, PostgreSQL, Bootstrap), requiring zero licensing fees, making it highly cost-effective to deploy on platforms like Render or Railway.

3.2 Users of the System

- **Learner (User):** Registers, reads topic content, attempts quizzes, participates in phishing simulations, and tracks personal scores.
- **Administrator:** Manages the question bank, oversees topic content, and monitors user engagement and scores.

3.3 Functional Requirements

1. **User Authentication & RBAC:** Secure registration and login using hashed passwords and session management.
2. **Topic Navigation:** Users must be able to view a list of topics and read detailed content for each.
3. **Quiz Engine:** The system must present multiple-choice questions linked to specific topics and calculate scores instantly.
4. **Progress Tracking:** The system must record when a user completes a topic and log their quiz scores permanently.
5. **Admin Dashboard:** Admins must have the ability to create, read, update, and delete (CRUD) quiz questions.
6. **Phishing Simulator:** An interactive module demonstrating safe vs. unsafe digital interactions.

3.4 Non-Functional Requirements

- **Security:** Passwords must be hashed. Routes must be protected by session guards. Admin routes must check for boolean flags.
- **Usability:** The interface must be mobile-responsive and visually engaging to encourage learning.
- **Reliability:** Relational ACID guarantees provided by the underlying SQL database.

3.5 Hardware & Software Requirements

4 System Design

4.1 Use Case Diagram

Figure 4.1 shows the primary actors (Learner User and Administrator) and their operational interactions within the application boundary.

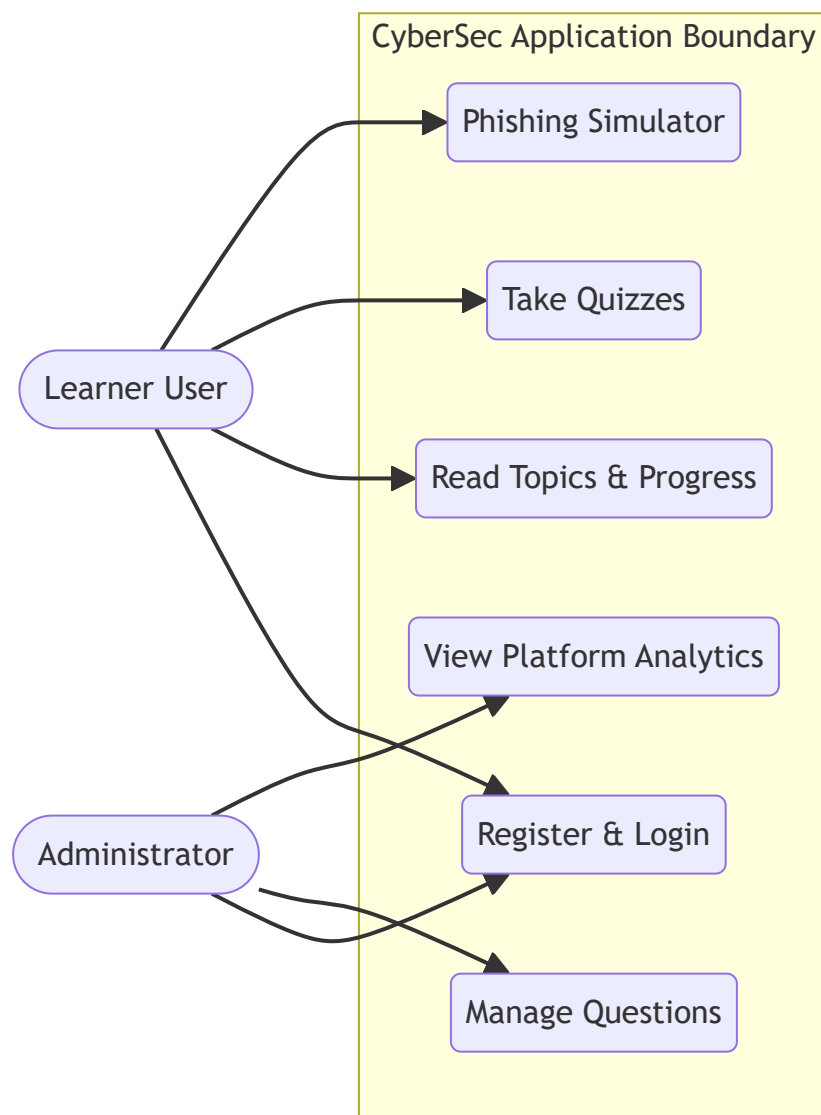


Figure 4.1: Use Case Diagram

4.2 System Architecture

Figure 4.2 illustrates the Model-View-Controller (MVC) deployment topology of the Flask application.

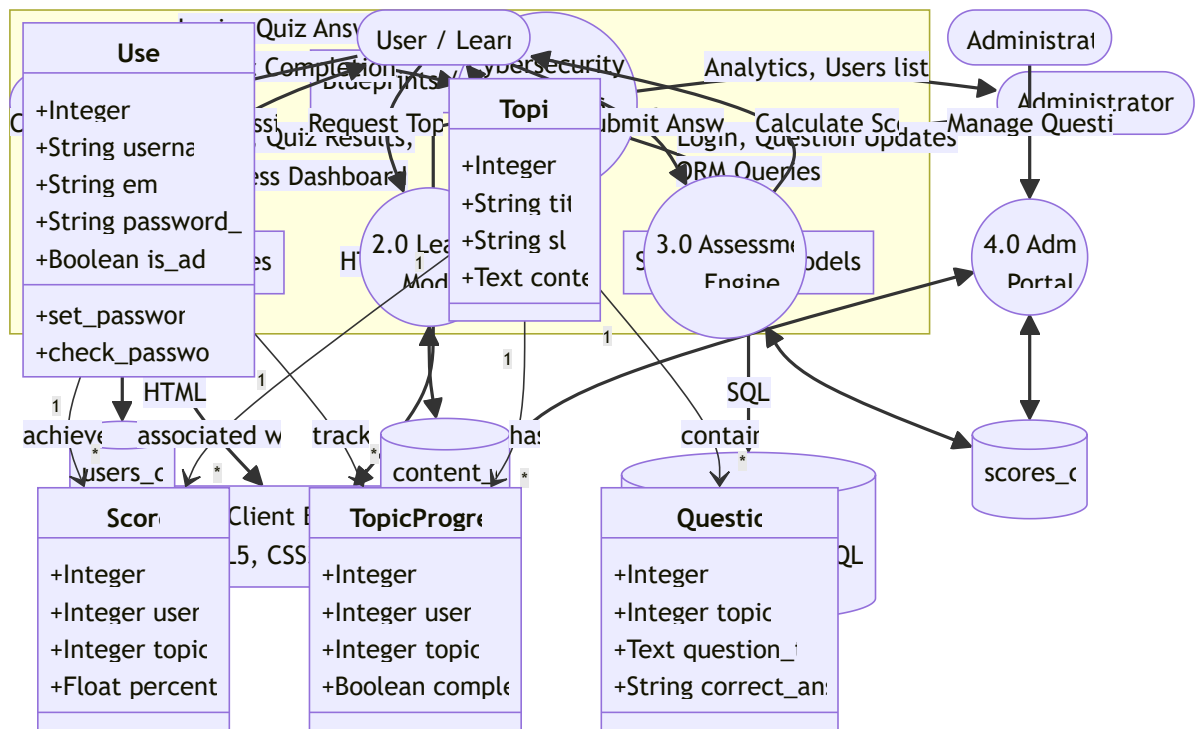


Figure 4.2: System Architecture

4.3 Data Flow Diagram (Level 0 – Context Diagram)

Figure 4.3: DFD Level 0

4.4 Data Flow Diagram (Level 1 – Subsystems)

Figure 4.4: DFD Level 1

4.5 UML Class Diagram

Figure 4.5: UML Class Diagram

4.6 Entity-Relationship Diagram

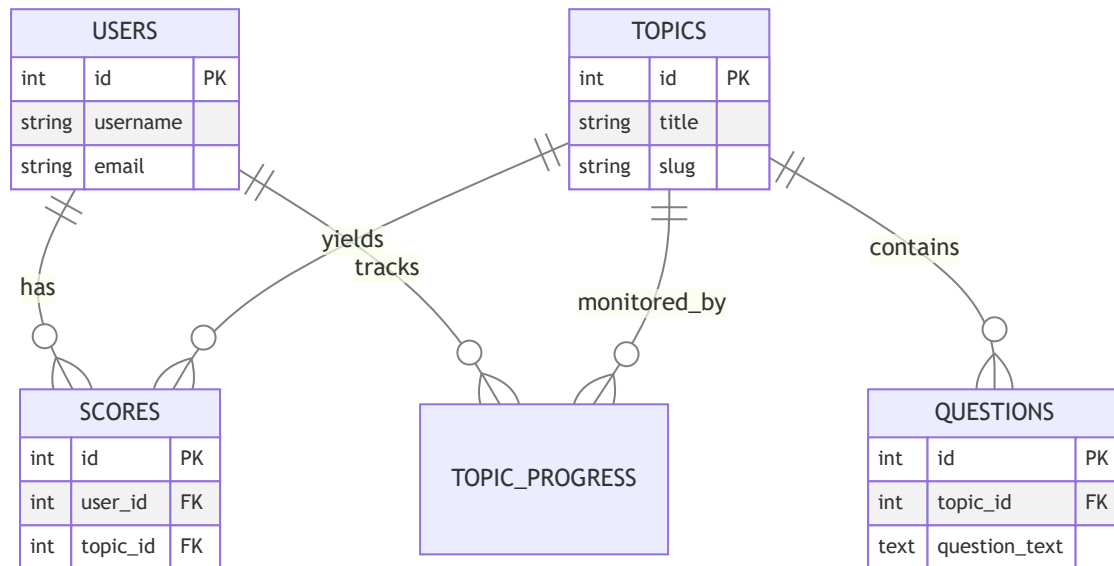


Figure 4.6: ER Diagram

4.7 Data Dictionary

Part III

Implementation & Testing

5 Implementation

5.1 Coding Convention

- **Modular Blueprints:** The application is divided into logical modules (`auth.py`, `main.py`, `quiz.py`, `admin.py`) using Flask Blueprints to keep routing clean and maintainable.
- **Environment Configuration:** Sensitive credentials and database URIs are loaded via `.env` files using `python-dotenv`, adhering to 12-Factor App principles.
- **ORM Data Management:** Direct SQL queries are avoided. All database interactions utilize SQLAlchemy objects for security and cross-database compatibility.

5.2 Security Measures

- **Authentication:** Sessions are managed securely using `Flask-Login`.
- **Password Storage:** Passwords are cryptographically hashed using `werkzeug.security.generate_password_hash` (`pbkdf2:sha256`). Plaintext passwords are never stored.
- **Access Control:** `is_admin` boolean flags ensure standard users cannot access the `/admin` routes.
- **Injection Prevention:** SQLAlchemy ORM inherently parameterizes queries, preventing SQL Injection attacks.

6 Testing

6.1 Test Strategy

Testing focuses on route authorization, database relational integrity (e.g., ensuring a score is tied to the correct user and topic), and accurate calculation of quiz percentages.

6.2 Test Cases

Part IV

Conclusion

7 Future Enhancement

- **Certificate Generation:** Automatically generate and email a PDF certificate upon completing all topics with a score above 80%.
- **Gamification:** Introduce leaderboards and achievement badges to increase user engagement.
- **Active Directory/SSO Integration:** Allow corporate users to log in using Google Workspace or Microsoft Azure AD.
- **Advanced Phishing Sandbox:** Interactive email inbox simulator where users must flag suspicious emails within a time limit.

8 Conclusion

The Cybersecurity Awareness Training Website successfully replaces static security guidelines with a dynamic, measurable, and engaging learning ecosystem. By utilizing a robust Python Flask backend and an intuitive Bootstrap frontend, the platform delivers critical security education efficiently. The integration of role-based access, automated quiz grading, and a relational database ensures that the application is highly scalable, secure, and ready to be deployed as an organizational training asset. This project demonstrates a practical application of full-stack web development principles in solving real-world digital security challenges.

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